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An experimental comparison of different kinds of laryngeal muscle reinnervation

HONGLIANG ZHENG, MD, SHUIMIAO ZHOU, MD, SHICAI CHEN, MD, ZHAOJI LI, MD, and YI CUAN, MD,
Shanghai and Dalian, China

In this study, we attempted to determine which method was the best for reinnervating the laryngeal adductor muscles by comparing nerve suture, nerve implantation, and nerve-muscular pedicle (NMP) transfer, as well as the length of time that could elapse after denervation and still allow for successful reinnervation with the ansa cervicalis. Reinnervation was performed in 36 dogs, at 6-, 8-, 10-, 12- and 18-month intervals after denervation via the three methods of muscle reinnervation described above. We noted some return of adduction in the cases using nerve suture before a 10-month interval after denervation, and with nerve implantation and NMP transfer before the 8-month intervals. The variable adduction was caused by reinnervation of the adductor muscles from the ansa cervicalis, as demonstrated by laryngeal spontaneous and evoked electromyography , the strength of muscle contraction, and histologic findings. Adduction was not observed in the cases after the above-mentioned intervals but partial improvement of the bulk and strength of the reinnervated vocal cord was still achieved. An analysis of the experimental results showed that nerve suture was superior to nerve implantation and the NMP technique. Little difference was noted between nerve implantation and the NMP technique. (Otolaryngol Head Neck Surg 1998;119:540-7.)

From the Department of Otolaryngology, Changhai Hospital, Second Military Medical University, Shanghai (Drs. Zheng, Zhou, Li, and Cuan); and Military No. 210 Hospitals, Dalian (Dr. Chen).
Reprint requests: Hongliang Zheng, MD, Room 301, Apt. 7, 500 Changhai Road, Shanghai 200433, P.R. China.
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0194-5998/98/$5.00 + 0 **23/77/80232**

In unilateral vocal cord paralysis (UVCP), the main problem is loss of active adduction during phonation. The rational purpose for the treatment of UVCP is to reestablish the adduction of the vocal cord during phonation through reinnervation of the adductor muscles. The literature on laryngeal reinnervation is extensive. Excellent reviews of the subject were published by Crumley,[1] Rice,[2] and Green et al.[3] What would be needed to reinnervate the adductor muscles is a nerve that activates during phonation and is anatomically adjacent to the larynx. Many investigators believed that the ansa cervicalis best fit these criteria.[3-6]

Three methods are available to reinnervate the laryngeal muscles using the source of the nerve: nerve suture (NS), nerve implantation (NI), and the nerve-muscular pedicle (NMP) technique. In 1924, the descendens hypoglossi anastomosis to the recurrent laryngeal nerve (RLN) was performed.[2] Recently, anastomosis of the branch of the ansa cervicalis to the RLN was proposed by Crumley and Izdebski[7] with satisfactory improvement in the voice of the patients. In our previous experiment, anastomosis of the main branch of the ansa to the adductor branch of the RLN could produce successful functional adduction during whining in dogs.[5] In the early 1970s, the ansa cervicalis NMP technique, as pioneered by Tucker et al.,[8] produced successful functional outcomes in laryngeal paralysis. It has been used by other workers with varying results,[1,9] prompting Tucker to publish detailed advice on the use of the NMP technique and the results that may be expected.[10] Paralyzed muscles may be reinnervated by implanting the cut end of a nerve.[11] Nerve implantation has been reported in the larynx using the RLN and the phrenic nerve. Our recent work[12] has confirmed that the subbranch of the main branch of ansa implantation into the adductor muscles may produce functional reinnervation.

In the current study, the purpose was to evaluate the chances for functional recovery of denervated laryn-

geal adductor muscles by NS, NI and the NMP technique, and to determine the length of time that could elapse after denervation and yet permit successful reinnervation by each of these methods of reinnervation.

METHODS AND MATERIAL

Experimental Design

Thirty-six adult dogs weighing between 10 and 15 kg were divided randomly into three groups: the NS, NI, and NMP technique groups. Under intraperitoneal pentobarbital anesthesia (40 mg/kg), 2.5 cm segments of the RLN and the external branch of the superior laryngeal nerve were removed to paralyze the vocal cord on the left side and both proximal and distal stumps were ligated in all animals. Three methods of reinnervation were performed respectively at the time of denervation and at 6-, 8-, 10-, and 18-month intervals of chronic denervation (two cases for each interval subgroup, 36 cases in total). Direct or videolaryngoscopy, spontaneous and evoked electromyography (EMG) were done before delayed reinnervation. Six months after the three methods of reinnervation, videolaryngoscopy, laryngeal spontaneous and evoked EMG, and contractile tension determinations were performed in the 28 living cases. At the end of the functional assessment, the thyroarytenoid (TA) and lateral cricoarytenoid (LCA) muscles in each case were taken for histologic studies. Eight of the dogs in the delayed reinnervation of the three methods did not survive the long interval but died of causes unrelated to the reinnervation procedure, including three, two, and three dogs, respectively, in the NS, NI, and NMP groups (Table 1). The work was carried out according to the guidelines of the Chinese Council on Animal Care for humane treatment of animals and approved by the Changhai Hospital Council on Animal Care.

Table 1. Adduction of reinnervated vocal cords

Group	Interval /months	Dog	Visible adduction	Stimulative adduction
NS	0	1	++	++
	0	2	+	++
	6	3	+	++
	8	4	+	++
	10	5	±	+
	10	6	+	++
	12	7	–	+
	12	8	–	+
	18	9	–	+
NI	0	10	+	++
	0	11	±	+
	6	12	+	++
	6	13	±	+
	8	14	+	++
	8	15	+	+
	10	16	–	+
	10	17	–	+
	12	18	–	+
	18	19	–	+
NMP	0	20	+	++
	0	21	+	++
	6	22	±	+
	8	23	±	+
	8	24	+	++
	10	25	–	+
	10	26	–	+
	12	27	–	+
	18	28	–	+

Results—excellent (++); satisfactory (+); resting in adducted position (±); and nonadduction (–).
NS, Nerve suture; *NI,* nerve implantation; *NMP,* nerve-muscle pedicle transfer.

Surgical Techniques

With the dogs under general anesthesia, a midline vertical incision was made in the neck. The loop of the ansa cervicalis on the left side was identified. It was found that the ansa cervicalis has a main branch that possesses two to four subbranches and innervates the inferior parts of the sternohyoid and sternothyroid muscles.

Nerve suture. The thyroid cartilage was engaged with a skin hook and gently rotated anteriorly. The inferior part of the cricopharyngeal muscle was cut and identification of the terminal branches of the RLN was carried out under an operating microscope. The adductor branch was transected at the bifurcation. The main branch of the ansa cervicalis was also sectioned near its first subbranch origin and anastomosed to the adductor branch of the RLN with four or five epineurial stitches of nylon 11-0 thread.[5] The wound was closed in layers.

Nerve implantation. The subbranches of the main branch of the ansa were dissected out and cut near their entry into the muscles. The cut ends were then laid carefully aside. The posterior, inferior half of the thyroid cartilage ala was removed, leaving an intact strut both posteriorly and inferiorly as described by Tucker.[10] The TA and LCA muscles were exposed. The epimysium was cut in the longitudinal direction at the middle segments of the muscles and one to two pockets in the TA muscle and one pocket in the LCA muscle were dissected. Then the proximal stumps of the subbranches of the main branch were implanted into these pockets. The epineurium was fixed to the epimysium by a 9-0 nylon suture to prevent dislocation of the nerves out of the pockets.[12] The outer perichondrium was then sutured back into place and the wound was closed in layers.

Nerve-muscular pedicle transfer. The subbranches of the main branch along with 2 mm square portions of muscles surrounding their terminal ends were harvested. The TA and LCA muscles were exposed and the NMPs were sutured into similar places cut out of the recipient muscles using 9-0 nylon sutures as described by Tucker,[8,10] with one to two pedicles for the TA muscle and one pedicle for the LCA muscle. In all other respect, the operation was similar to the NI procedure.

Videolaryngoscopy

Videolaryngoscopy was carried out on very lightly anesthetized dogs to detect the degree of spontaneous motion of

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the operated cords during laryngeal stimulation to whine and obstructive respiratory dyspnea. It was also carried out during single nerve stimulation of the nerve transfer. The image was detected by videocassette recorder. In all the living animals, the strap muscles and cricothyroid muscles on both sides were denervated during data collection to ensure that the vocal cord movements observed were not caused by these muscles.[1,4]

Electromyography and Contractile Tension

A four-channel electromyograph and bipolar needle electrodes (Dantec Counterpoint, Copenhagen, Denmark) were used for the EMG recordings. Spontaneous EMGs in the TA and LCA muscles were performed during whining under normothermic conditions (37° C). Evoked EMGs of the TA muscles were performed by stimulating proximal to the site of transection on the left side before delayed reinnervation, and the main branch of the ansa on the operated side and the RLN on the normal side 6 months after reinnervation in each case at the same distance from the TA muscle, with a single 0.1 msec pulse of sufficient voltage to obtain an adequate response. The amplitudes of spontaneous and evoked EMGs both before and after the operation in each left TA muscle is reported in relative terms by using the unoperated control TA muscle as a percent of the corresponding value for the right side. The average of these values was calculated for each interval subgroup. Contractile tension of the TA muscle was determined as described by Rice and Cooper.[13] After EMG examination, the TA muscle was attached to a force transducer that was connected to a chart recorder. Then the left transferred nerve and the right RLN were stimulated supramaximally for twitch contraction with a single 0.1 msec pulse (1 pulse/sec) and for a maximal tetanic contraction with 100 Hz. These two values are reported in relative terms, as done in the amplitude of the EMG.

Histologic Studies

After completion of the physiologic studies, both the right and left TA and LCA muscles were removed. Acetylcholinesterase activities at the motor end plates (MEP) and the nerve fibers were studied with the cholinesterase-Bielschowsky staining method in these muscles. The muscles were also treated with hematoxylin and eosin (HE) stain.

RESULTS

Videolaryngoscopy and Macroscopic Findings

The left vocal cord rested in the paramedian position in all cases before reinnervation and appeared atrophic to various degrees in the cases with delayed reinnervation. Five dogs before the 10-month interval for reinnervation in the NS group, and 4 and 3 dogs before the 8-month interval in the NI and NMP groups, respectively, were observed to achieve a variable degree of adduction of the vocal cord on the left side during whining and were noted to abduct to the paramedian position after adduction. Immediate silence occurred after the section of the transferred nerve. Except for slight contraction in dog 10, no adduction was noted during obstructive respiratory dyspnea in these cases. In dogs 5, 11, 13, 22, and 23 the reinnervated vocal cord was observed to be in the fully adducted position at all times. In the above-mentioned cases, the left cord did not appear atrophic. In the other 11 cases, the left cords appeared to rest in the paramedian position but they were noted to adduct to variable degrees with stimulation of the transferred nerve (Table 1). The bulk of the cords increased compared with that before reinnervation in all the delayed reinnervated animals.

Electromyographic Findings

Complex, repetitive discharges could be recorded in the denervated adductory muscles in 6 cases before delayed reinnervation. The other 16 cases showed regular motor unit potentials that were low (Fig. 1), showing single or partial interference patterns. The amplitude of the evoked EMG could be recorded in these cases, but it was less than 2% to 8% of the unoperated side.

Six months after reinnervation, the spontaneous electrical activities of the left TA and LCA muscles could be recorded in all cases during whining, which was synchronous with the discharge of the normal TA muscle. The EMGs in the cases before the 10-month interval in the NS group and before the 8-month interval in the NI and NMP groups showed normal interference patterns. They showed partial interference patterns in the other cases. The amplitude of the motor unit potential on the reinnervated side was found to be less than that on the normal side in most of the animals (Fig. 1, Table 2). On average, the difference in the EMG amplitude between the NI and NMP groups was not significant at the $p > 0.05$ level. The EMG amplitude of the NS group was significantly greater than those of the NI ($p < 0.05$) and NMP ($p < 0.01$) groups.

The amplitude of the evoked EMG in the reinnervated TA muscle is shown in Table 2. It was noted that the amplitude appeared to vary inversely with the period of denervation. On average, there was a significant difference in the evoked EMG amplitude between the NS and NI groups ($p < 0.01$), and between the NS and NMP groups ($p < 0.01$). No significant difference was shown between the NI and NMP groups ($p > 0.05$).

Measurement of Muscle Tension

Table 3 shows the strength of contraction in relative terms. As can be seen from the table, the twitch and

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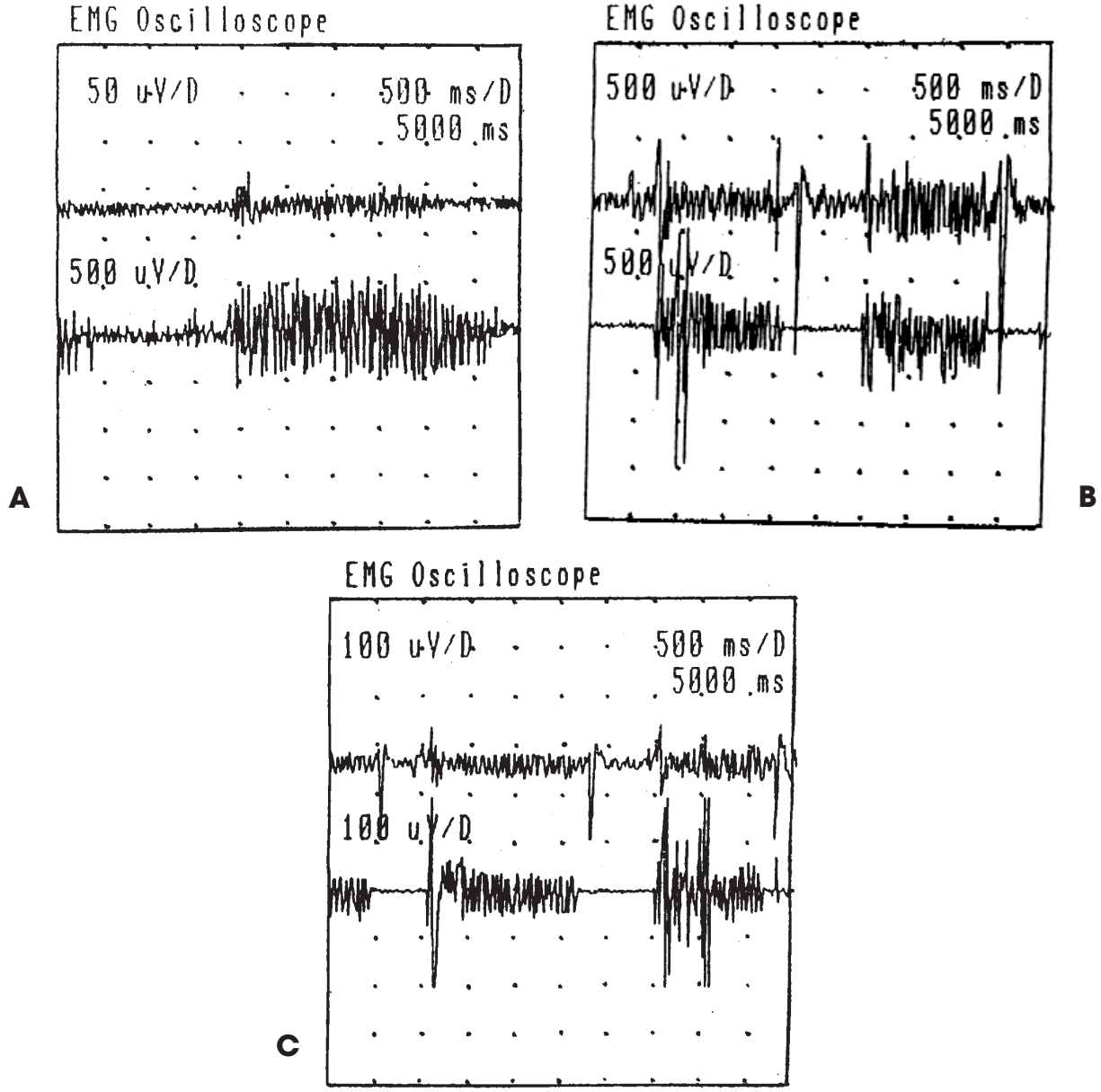


Fig. 1. Spontaneous EMG. Electrical activities of the left TA muscle *(upper trace)* during phonation were synchronous with those of the normal side *(lower trace)*. **A,** Electrical activities of left preoperative TA muscle (dog 14, 8-month interval for denervation) could be recorded. **B,** Electrical activities of delayed reinnervated TA muscle (dog 14, 8-month interval for NI) were a little lower than those of the normal side.**C,** Electrical activities (dog 27, 12-month interval for NMP technique) were obviously lower than those of the normal side.

tetanic tension decreased gradually with the period of denervation. On average, a significant difference in the twitch and tetanic tension was observed between the NS and NI groups ($p < 0.01$). The difference between the NI and NMP groups was not significant at the $p < 0.05$ level.

Histologic Findings

Only slight signs of atrophy and fibrosis were noted in the reinnervated TA and LCA muscles before the 10-month interval for reinnervation in the NS group, and in the 8-month interval in the NI and NMP groups. The muscle fibers appeared nearly normal. The MEPs were arranged in good order and occupied about two thirds

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Table 2. The amplitudes of spontaneous and evoked EMG

Interval of denervation (months)	Amplitude of EMG (%)			Amplitude of evoked EMG (%)		
	NS	NI	NMP	NS	NI	NMP
0	98.8	63.4	79.3	92.1	67.6	64.2
6	89.7	75.1	74.2	78.3	58.1	53.8
8	70.3	64.2	63.3	68.3	47.8	49.2
10	68.3	59.6	54.6	60.4	42.7	37.6
12	53.7	40.7	36.5	38.5	23.6	21.4
18	44.2	34.0	23.8	28.2	14.5	16.8

NS, Nerve suture; *NI,* nerve implantation; and *NMP,* nerve-muscular pedicle transfer.

Table 3. Strength of contraction (%)

Interval of denervation (months)	NS		NI		NMP	
	Twitch	Tetanic	Twitch	Tetanic	Twitch	Tetanic
0	96.3	102.4	79.0	81.6	72.3	83.4
6	85.1	92.7	63.2	76.1	59.3	67.1
8	72.8	70.5	58.2	61.8	54.4	56.9
10	60.5	68.4	46.7	58.9	44.5	52.1
12	47.2	54.6	33.6	42.1	39.0	36.7
18	23.3	25.2	14.3	16.7	12.4	15.0

NS, Nerve suture; *NI,* nerve implantation; and *NMP,* nerve-muscular pedicle transfer.

of the muscles length. The shape of the MEPs was uniform, and new nerve fibers entering into the MEPs could be noted. The fine structure of the MEPs could be distinguished in some lightly stained specimens (Fig. 2). After the above-mentioned intervals, the MEPs were obviously disordered, decreased, and distributed in a widely scattered manner on all the TA and LCA muscles. The sprouting nerve fibers could be found to extend long distances between muscle fibers and to reestablish new neuromuscular junctions in some places (Fig. 3).

Mature and regular MEPs were present in larger numbers in the NS group than in the NI and NMP groups at the same intervals for delayed reinnervation. Little difference was observed between the NI and NMP groups.

DISCUSSION

Because not all nerve axons have the regenerative power to establish neuromuscular junctions, the transferred nerve must have a sufficient number of axons to reach the paralyzed muscle and produce adequate reinnervation for muscular contraction.[1] In our previous studies, the main branch of the ansa cervicalis, which has more fibers than the branches of the ansa,[14] was used to immediately reinnervate the adductor muscles by NI[12] and NS.[5] The results demonstrated that the main branch of the ansa seemed to produce satisfactory reinnervation of the adductor muscles.

In the current study, we attempted to determine which of the three immediate and delayed reinnervation procedures is best for reinnervating the adductor muscles. This is of clinical importance in treatment of UVCP if reinnervation is considered. Comparing the results of the different kinds of reinnervation in the TA muscles, we found that NS was superior to NI and NMP transfer. Nevertheless, NI and NMP transfer were successful in all cases and could lead to functional recovery with variable degree. Sometimes the interpretation of the EMG and videolaryngoscopy findings in a reinnervated vocal cord had some problems, but together with the strength of muscle contraction and histologic findings, the final assessment was reliable.

These results support former experimental findings that NS is the best method to reinnervate skeletal muscles[15] and transfer of a neuromuscular pedicle is an effective reinnervation method.[6,10] In addition, we have shown that a similar result may be obtained by simple implantation of the cut end of a nerve, which is in agreement with the findings of other authors.[11] Multinerve or multipedicle implantations in the NI and NMP groups seemed to produce satisfactory reinnervation. It is difficult, however, to compare the results achieved with reinnervation by the main branch of the ansa cervicalis to those achieved by a single nerve transfer because they were not compared in this study. What is the difference between NMP transfer and NI? This study has shown that both procedures are effective and the results are not significantly different. In fact, a 2 mm cube of muscle obtained by the method of Tucker might contain cut ends of the nerve and so behave more like a nerve implant.[11] If this is so, there may be little difference in practical terms between an NMP transfer and NI.

Reinnervation must take place during a poorly delineated time period before the muscle fiber undergoes atrophy from denervation. Denervation atrophy can be prevented only by reinnervation.[1] The length of time for this atrophy to occur is not clear. Cutamn and Young[1] have shown that even after 1 year, successful reinnervation can still be achieved in skeletal muscle. Even longer periods have been shown with regard to facial muscles,[1] but the successful reinnervation of long-term denervated laryngeal muscles has rarely been investigated in controlled animal experiments. Morledge et al.[16] concluded that posterior cricoarytenoid (PCA) muscle reinnervation was feasible 2 to 6 months after denervation, but not after 9 months had elapsed. This comment was not strongly supported by

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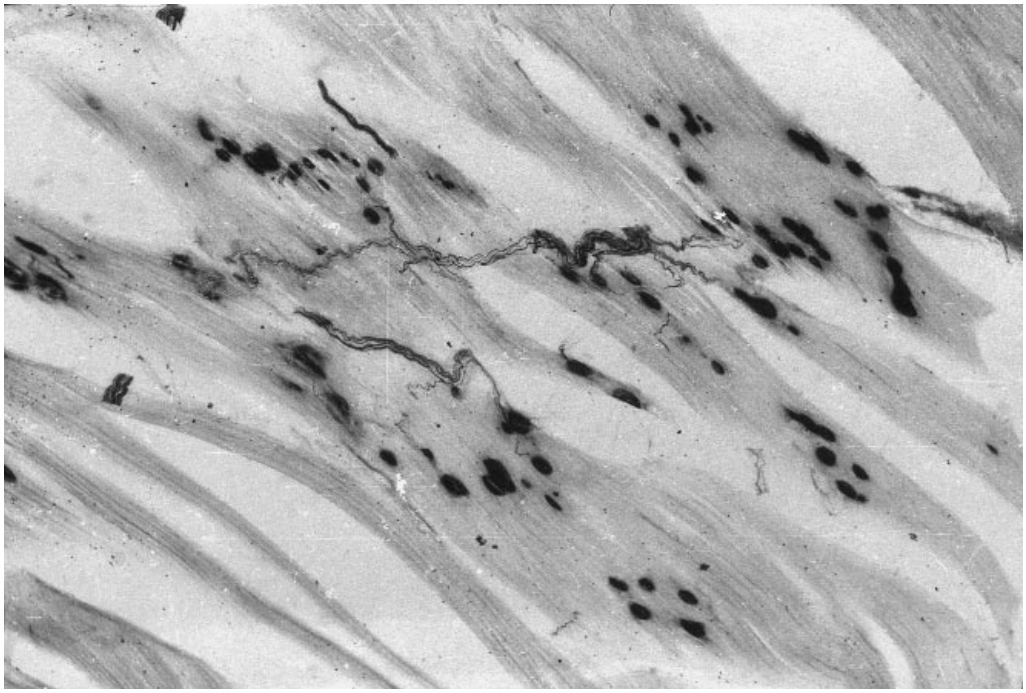


Fig. 2. The MEPs and nerve fibers of reinnervated TA muscle with NMP technique (dog 24, 8-month interval for reinnervation), showed mature and regular MEPs, and new nerve fibers entering into the MEPs. (Original magnification × 20.)

their data. Lyons and Tucker[6] and Neal et al.[9] reinnervated the long-term denervated PCA muscles with the NMP technique. They found that restoration of abduction could be achieved in animals waiting 6 to 8 months for reinnervation.

To our knowledge, laryngeal delayed reinnervation 12 months after denervation has not been reported in controlled animal experiments. In the current study, some return of adduction in the cases before the 10-month interval of chronic denervation in the NS group and before the 8-month interval in the NI and NMP groups could be achieved. It has been noted that the TA muscle tension and the amplitudes of spontaneous and evoked EMGs in these cases were more than or near 50% of the normal values. Reinnervation in the cases after the above-mentioned periods of denervation could still be achieved with variable tension and improved bulk of the left vocal cord.

Motor endplates are not amenable to reinnervation unless all those present in a "denervated" muscle have been completely denervated.[1,17,18] On the other hand, reinnervating axons exhibit a high probing activity and a special capacity for synapse formation at the former synaptic sites in denervated muscles.[19,20] In the current study, 2.5 cm segments of RLN and superior laryngeal nerve were removed to completely paralyze the laryngeal muscles. The study showed that the distribution of MEPs was nearly normal in the reinnervated muscles before the 10-month interval of denervation. This result can be accounted for if the regenerating axons could still form neuromuscular junctions at former junctional sites in the delayed reinnervated muscles. In addition, the MEPs were distributed in a widely scattered manner in all the TA and LCA muscles after a 12-month interval for reinnervation. The results support previously published research showing that denervated muscle fibers in general are capable of forming synapses at new sites during reinnervation.[19,21] We have noticed that scattered and immature MEPs appear to increase with the period of denervation before reinnervation. The function of the irregular and immature MEPs, however, was too low to produce sufficient muscular contraction.

Under clinical conditions, most of the original nerve supply is intact but nonfunctional in the patients with UVCP. It is customary to wait at least 6 months after onset of UVCP before performing the operation to allow for possible spontaneous reinnervation.[6] Our current study indicates that reinnervation by three different methods is still possible after a delay of 18 months. If, however, the NMP or NI procedures were considered under clinical conditions, a bar to successful reinnervation may occur because a small number of the RLN is

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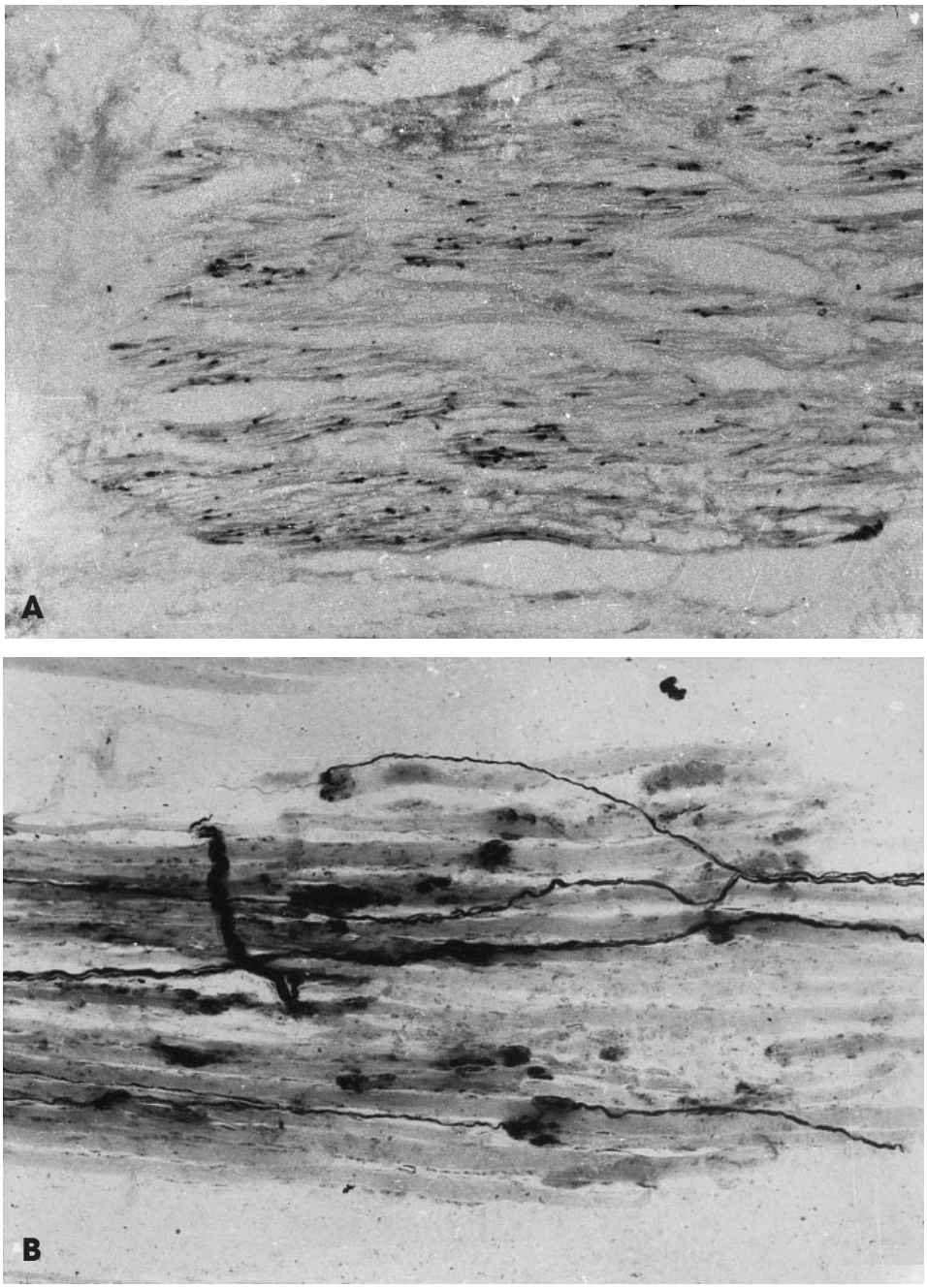


Fig. 3. MEPs and nerve fibers of reinnervated TA muscle with nerve suture (dog 9, 18- month interval for reinnervation). **A,** Disordered and scattered distribution of the MEPs. (Original magnification × 10.) **B,** Irregular and immature MEPs and extended nerve fibers. (Original magnification × 50.)

able to regenerate back to the laryngeal muscles and yield subclinical innervation, which inhibits the development of the newly implanted nerve or NMP.[1] In this situation, however, the adductor branch of the RLN or RLN[7] anastomosis to the ansa cervicalis is still suitable because the subclinical innervation was excluded after cutting the RLN or its adductory branch. We think that NS should be considered first if the patient with UVCP chooses reinnervation. Because NI or NMP transfer do not damage the recurrent laryngeal nerve, immediate reinnervation should be a possibility in clinical application. If these two procedures are performed in patients with long-term laryngeal paralysis, subclinical innervation should be avoided by cutting the RLN or its branch.

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CONCLUSIONS

Analysis of an experimental study on reinnervation of the paralyzed laryngeal muscles leads to the following conclusions:

1. Nerve suture is superior to NI and NMP transfer in the treatment of long-term laryngeal paralysis.
2. Nerve implantation and NMP transfer are successful and could lead to functional recovery with variable degree.
3. No significant difference was observed in the reinnervation obtained by NI or NMP transfer.
4. Reinnervation with the three different methods appears to vary inversely with the period of denervation.
5. Variable return of adduction could be achieved in the cases operated before a 10-month interval for reinnervation with NS and an 8-month interval with NI and NMP transfer.
6. The reinnervated vocal cord also showed improved volume and tension in the cases after the above-mentioned intervals for reinnervation.

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